SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	ASEA
Project / Product Name	Path.AI: find your path in the AI era
Final Live Deployment URL	https://pathai.lol
Codebase Repository URL	https://github.com/Almatikooo/Competition_Repo
Name (Leader)	Almat Sakiidulla
Email & phone (Leader)	sat200707@mail.ru, +77072203626 (WhatsApp)

2. Project Description

[Maximum 250 words]

Path.AI (https://pathai.lol) is a live web app for students, graduates and workers whose jobs are changing with AI. It answers one question: "What should I learn next, and for which job?"

A person enters their job or field of study, or uploads a CV, and confirms the skills found. A transparent engine ranks 43 future-facing roles by skill fit, transferable strengths, demand under three AI-adoption scenarios, interests and SDG priorities. Each match explains the fit, names missing skills and estimates learning hours.

The plan for a chosen role orders the missing skills and links each to a free, open-licensed course inside Path.AI (53 courses with lessons, tests and verifiable certificates) and a checked external resource. Opportunities shows matching jobs from 19,481 postings and universities in 219 countries. Every role and skill is linked to UN SDG targets, led by 4.4.

An open-weight model, Qwen3.8-27B on our own RunPod GPU, reads CVs and job posts and writes reviews and AI Chat answers, using only facts the engine computed; any other number is rejected. The site works in English, Malay and Chinese.

New in Phase 2: sign-in with Google, Apple or GitHub keeps progress on every device; My path turns a role into a goal with readiness, a roadmap, weekly hours done and left and a daily streak; 100 achievements earn points toward real gifts. Our survey of 55 young people confirms the need: 53% found choosing a profession hard and 51% struggled to get a first job or internship.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Path.AI is one **FastAPI (Python)** service serving a no-build JavaScript single-page app (English, Malay, Chinese) and a JSON API, on a DigitalOcean VPS in Singapore behind Caddy (HTTPS); it also runs on Replit.

Engine. Deterministic code computes every number: skill fit with credit for transferable skills, demand under three AI-adoption scenarios, and an ordered pathway with hours and weeks.

GenAI. Qwen3.8-27B-FP8 (open weights) on vLLM, RunPod RTX A6000, OpenAI-compatible API; started on demand, stopped after 20 idle minutes. Backup: the same model on Groq (never sent a CV); fallback: 153 saved answers. Guards drop CV skills without a verbatim quote and reject AI text with numbers not in the computed facts.

Data and identity. Dated JSON catalogs and caches in Git; SQLite for accounts. New: Google (OAuth 2.0 + PKCE), GitHub and Apple sign-in, with HMAC-signed sessions bound to a revocable per-account epoch.

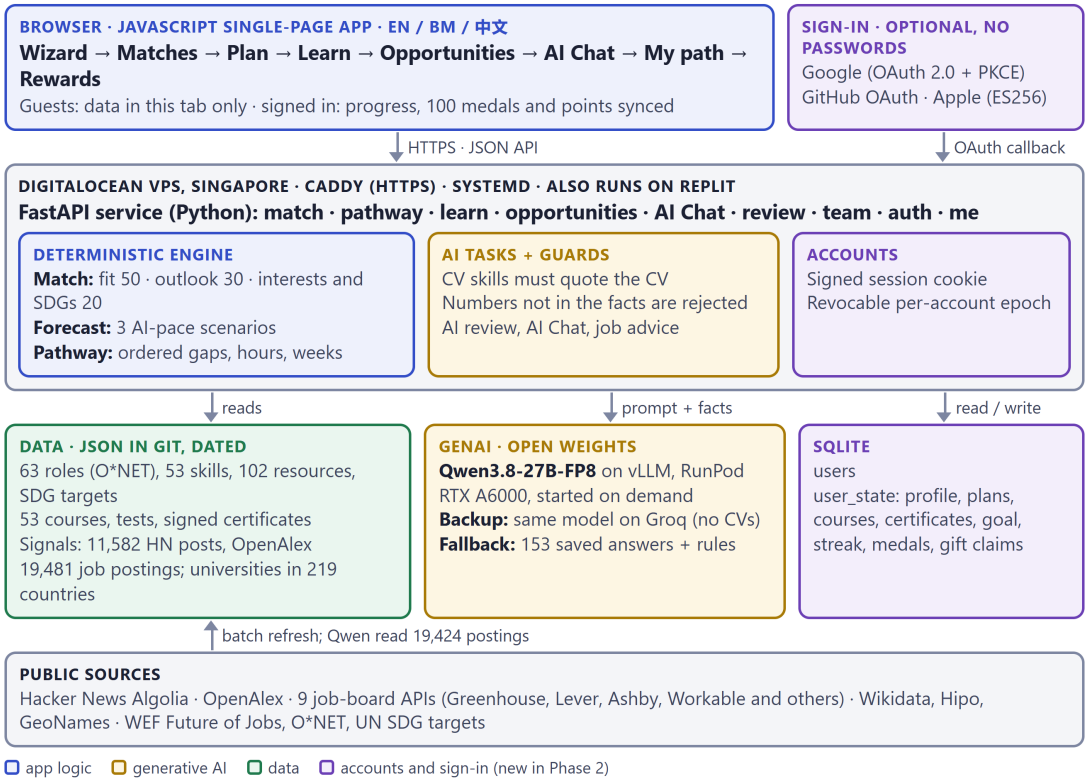


Figure 1. Final architecture as deployed at https://pathai.lol (commit 7907110, 30 October 2026). Purple parts are new in Phase 2.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Built and deployed for Phase 2 (16 commits, 28–30 Oct 2026; 162 tests passing):

- **Accounts without passwords.** Sign-in with Google, GitHub and Apple, a fourth wizard step to save results or continue as a guest, a My profile page and progress synced to SQLite. In Phase 1, progress lived in one browser only.
- **My path.** A chosen role becomes a goal: readiness % with its trend, a roadmap following real course progress, "about N weeks to go", time learned, weekly-goal hours done and left and a 12-week activity calendar.
- **Daily streak.** Daily visits grow a streak shown in the header with the best streak; one missed day is forgiven, two end it. Streak medals unlock at 2, 3, 5, 7, 14, 30 and 100 days.
- **Achievements and rewards.** 100 medals (70 bronze, 25 silver, 5 gold) for learning, tests, certificates, streaks and goals, announced with a chime. They earn points (10, 50 or 200; 1 point ≈ RM 1) that pay up to half of one of 30 gifts, from stickers to a MacBook Air (demo only).
- **More professions.** 63 roles and 53 skills, up from 35 roles and 44 skills in our Phase 1 proposal, and 53 free courses.

Pivots from the Phase 1 design

- **Stack.** We planned React, TypeScript, Tailwind and PostgreSQL but shipped a no-build JavaScript app, JSON catalogs and SQLite only for accounts. It loads fast on mobile networks, runs on a 1 GB server, and every data change is reviewable in Git.
- **Privacy.** Sign-in is optional: guests' data stays in the open tab only, accounts store no passwords, and deleting an account deletes all its data.
- **Focus.** Many respondents struggle to choose a profession, so we built tools that bring learners back daily.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	**SDG targets Path.AI serves** (official UN wording in quotes): **SDG 4.4 (primary): relevant skills for decent jobs.** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Path.AI finds a person's skills, names the missing ones for a future-facing role and turns them into an ordered plan with hours, free courses, tests and verifiable certificates. The benchmark plans need 68 to 234 learning hours, scheduled at each person's weekly pace. **SDG 4.3: affordable technical, vocational and tertiary education.** "By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university." All 53 in-app courses are free and open-licensed, and all 23 benchmark plan steps start with a free (14) or free-to-audit (9) resource. Opportunities → Study lists universities in 219 countries, with official admission and scholarship portals for popular destinations. **SDG 8.5: productive employment and decent work.** "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value." Path.AI matches the profile with 19,481 real job postings whose requirements our model read from each posting (26 hand-checked postings: degree 96%, pay 96%, skills recall 92%). Rankings ignore age, gender, ethnicity and location, and a test enforces this. **SDG 8.2: productivity through technological upgrading.** "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Path.AI estimates skill demand under slower, expected and faster AI adoption from 11,582 job posts and OpenAlex research, and moves people from tasks AI automates to roles that use it. All 6 benchmark top roles stay in the top 5 under all three scenarios. **Phase 1 targets against the final prototype:** • **Complete plans.** Target: 100% of the 6 required elements. **Achieved: 100%.** All 6 benchmark profiles get a role, an explanation with sources, skill gaps with hours, ordered steps, resource links and SDG links; 23 of 23 steps also have a free in-app course.

Metric Category	Your Response
	• **Valid links.** Target: 100%. **Achieved: 100%**, 114 of 114 learning-resource, portal and source URLs, 0 broken (30 Oct 2026). • **Planning time.** Target: at least 50% less median time than manual research (provisional baseline 30 min, so 15 min or less). System time: 0.08 s per profile (match + plan over the internet).
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	• Script against the live API (code and log in 4.1): 6 benchmark profiles, every element checked. • scripts/verify_resources.py requests every learning-resource, support-portal and cited-source URL. • 162 pytest tests on the final commit, including a fairness test (age, gender, ethnicity and location cannot change rankings), the AI number guard, sign-in and certificates. • Dated data in the repository: 11,582 HN posts (2023-11 to 2026-10), OpenAlex 2024H1–2026H1, 19,481 job postings from 213 sources. • Stopwatch sessions with the Phase 1 personas and the same checklist; survey of 55 people (22–25 Oct 2026).
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	In our survey, 53% found choosing a profession hard, 51% struggled to get a first job or internship and 27% know little about the skills AI-era jobs need. Path.AI replaces guesswork with a free, sourced plan: 53 open-licensed courses (SDG 4.3), verifiable certificates whose link people add to a CV or LinkedIn (SDG 8.5), and streaks, 100 achievements and gift rewards that help people finish what they start. **Partners scale it:** career centres give every student a plan and see cohort readiness; employers plan reskilling in the team (HR) view; agencies such as HRD Corp or MYFutureJobs can add their courses and vacancies; companies can sponsor the reward gifts.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	**In production** at https://pathai.lol on a 1 GB VPS; the GPU is billed only while it runs. Every feature is a JSON API (match, pathway, learn, opportunities, certificate check), so a university portal or HR system can embed Path.AI as a plug-in service. **Planned revenue (from our survey):** a free tier as the funnel; Plus at 4,990 KZT (≈ $11) with AI review and chat, internship matching and all certificates; Premium at 9,990 KZT (≈ $22) for career changers; annual licences for universities and employers. 93% of respondents would pay at least 5,000 KZT for one of the two tools we tested (87% for AI-skills learning with internship matching); a pre-order page will test real conversion.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

1. Core KPI from Phase 1: time to a complete career plan

$$\text{Time reduction (\%)} = (T_{manual} - T_{Path.AI}) \div T_{manual} \times 100$$

Target, unchanged from Phase 1: at least 50%. With the provisional baseline T_{manual} = 30 min, Path.AI must take 15 min or less, and a plan counts only when all six required elements are present (part 4).

2. How a role is scored: a worked example (src/app/engine/matching.py)

Every number on screen comes from this deterministic formula; the language model only explains it. Default weights, which the user can change:

$$\text{Overall} = 0.5 \times \text{Fit} + 0.3 \times \text{Outlook} + 0.2 \times \text{Preference}$$

$$\text{Fit} = \Sigma\ (w_i \times \text{credit}_i) \div \Sigma\ w_i$$

$$\text{Outlook} = \text{clamp}(50 + 250 \times \text{growth}, 0, 100)$$

Credit for a skill by level: none 0, basic 0.4, working 0.75, advanced 1.0; a related, transferable skill gives at most 0.5. Growth is the role's estimated yearly change in demand in the chosen AI-adoption scenario. Preference is the overlap with the person's interests and SDG priorities.

Example: Aisyah, an accounts clerk, and the role Business Analyst at the expected AI pace:

$$\text{Fit} = 3.525 \div 5.0 = 70.5\%$$

$$\text{Overall} = 0.5 \times 70.5 + 0.3 \times 56.9 + 0.2 \times 75.0 = 67.3 \ \rightarrow\ \text{rank } 1$$

Analytical thinking, communication and Excel count fully; data analysis counts 0.5 through her Excel; SQL, project management and two minor skills count partly. The role stays in the top 2 under the slower and faster scenarios too, so the advice holds if AI spreads faster.

3. Guard against invented numbers (src/app/llm/tasks.py)

An AI answer that contains any number not present in the computed facts is rejected, and the page shows rule-based text instead. Tests:
test_explanation_guard_rejects_invented_numbers_and_citations,
test_api_falls_back_to_template_when_model_invents_a_number.

```
def guard_numbers(text, facts):
    stray = numbers_in(text) - numbers_in(json.dumps(facts, ensure_ascii=False))
    return [f"numbers not in facts {sorted(stray)}"] if stray else []
```

4.1 Mathematical Proof / Code Logic Snippet (continued)

4. Complete plans on the live site, 30 October 2026

The script asks https://pathai.lol for the top role and the plan of each of the 6 benchmark personas and checks the six plan elements required in Phase 1.

```
import time, httpx
B, c = "https://pathai.lol", httpx.Client(timeout=60)
courses = {x["skill"] for x in c.get(B + "/api/learn/courses").json()["courses"]}
personas = c.get(B + "/api/personas").json()["personas"]
complete, steps_n, with_course, secs = 0, 0, 0, []
for p in personas:
    t = time.perf_counter()
    top = c.post(B + "/api/match", json={"profile": p}).json()["roles"][0]
    plan = c.post(B + "/api/pathway",
                  json={"profile": p, "role_id": top["id"]}).json()
    secs.append(time.perf_counter() - t)
    steps = plan["steps"]
    check = {"role": bool(top["id"]),
             "explanation": bool(top["explanation"] and top["sources"]),
             "skill_gaps": all(s["hours"] > 0 for s in steps),
             "ordered_steps": [s["n"] for s in steps] == list(range(1, len(steps) + 1)),
             "resource_links": all(s["resource"]["url"].startswith("https://")
                                   for s in steps),
             "sdg_links": len(plan["sdg_contribution"]) > 0}
    complete += all(check.values())
    steps_n += len(steps)
    with_course += sum(s["skill"] in courses for s in steps)
    print(f'{p["id"]:8} -> {top["id"]:28} {len(steps)} steps {plan["total_hours"]:>4} h'
          f' {plan["weeks"]:>3} wk  complete={all(check.values())}')
print(f"complete plans {complete}/{len(personas)} | steps with an in-app course"
      f" {with_course}/{steps_n} | mean time per persona {sum(secs) / len(secs):.2f} s")
```

Output:

```
aisyah   -> business_analyst             5 steps  122 h  21 wk  complete=True
weijie   -> ev_technician                2 steps   68 h   6 wk  complete=True
kavitha  -> customer_success_manager     3 steps   81 h  17 wk  complete=True
hafiz    -> ux_designer                  3 steps  161 h  21 wk  complete=True
nurul    -> precision_agri_technologist  5 steps  176 h  26 wk  complete=True
arif     -> automation_technician        5 steps  234 h  39 wk  complete=True
complete plans 6/6 | steps with an in-app course 23/23 | mean time per persona 0.08 s
```

Result: 6 of 6 complete plans, 23 of 23 steps with a free in-app course, 0.08 s per persona.

5. New in Phase 2: the daily streak rule (src/static/core.js)

A visit within two days of the previous one continues the streak: one missed day is forgiven, two end it. Example: Monday, Tuesday, Thursday → a 3-day streak; Monday, Thursday → 1 day.

```
if (run && d <= shiftDay(run.end, 2))
  Object.assign(run, { count: run.count + 1, end: d });   // one missed day: the run goes on
else runs.push({ count: 1, end: d });                     // two missed days: a new run
const alive = !!last && last.end >= shiftDay(today, -2);  // visited in the last 2 days
```

6. Links and regression tests on the final commit 7907110

```
$ python scripts/verify_resources.py    →   114 URLs checked, 0 broken
$ pytest -q                             →   162 passed, 1 warning in 32.98s
```

Checks every resource, portal and source URL; the tests cover matching, fairness, AI guards, sign-in.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Path.AI already runs in production, so scaling means more data, users and partners, not a rebuild.

Market. Our survey (55 responses, 22–25 Oct 2026, mostly students in Kazakhstan) shows students (60%) as the core users and working graduates (25%) as the most willing to pay. Matching, plans and courses stay free; Plus (4,990 KZT ≈ $11) and Premium (9,990 KZT ≈ $22) will be tested with a pre-order page, and universities and employers buy annual licences for career-centre and HR dashboards.

Real-world data at volume. With funding or grants we will:

1. move accounts from SQLite to managed PostgreSQL and run several app instances behind a load balancer (only the account database holds user data);
2. refresh signals and job postings daily and add Malaysian sources (MYFutureJobs, MyMahir, DOSM, HRD Corp) through data agreements;
3. read new postings in nightly vLLM batches (19,424 already read) and serve users from the dated cache, so a plan still takes a fraction of a second;
4. scale GPU capacity automatically, only for AI text.

Partnerships. Joint research at XMUM to calibrate our skill-sensitivity assumptions on observed Malaysian job-ad changes and to measure learning outcomes; Rotary and corporate partners to sponsor Premium seats and reward gifts for low-income learners and offer internships linked to plans; universities to pilot cohort dashboards.

Reach. Malay and Chinese course content, Russian and Kazakh interfaces, local salaries, and the survey repeated in Malaysia with ringgit prices.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. As declared in Phase 1, the repository was created on 19 October 2026 at 22:55 MYT, after the problem statements were released; all code, catalogs and lessons were written during the hackathon.

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	Code (web framework)	MIT
Uvicorn	Code (ASGI server)	BSD-3-Clause
Pydantic	Code (data validation)	MIT
httpx	Code (HTTP client)	BSD-3-Clause
python-multipart	Code (file upload)	Apache-2.0
pypdf	Code (CV PDF text)	BSD-3-Clause
pytest	Code (tests)	MIT
cryptography (new in Phase 2)	Code (Sign in with Apple ES256 key)	Apache-2.0 or BSD-3-Clause
vLLM	Code (model server)	Apache-2.0
Roboto (Google Fonts)	Code (font)	SIL OFL 1.1

Name	Type (code/dataset/model)	License
flag-icons	Code (country flags)	MIT
Qwen3.8-27B-FP8 (Alibaba Qwen)	Model, pre-trained (vLLM on RunPod; Groq as backup)	Apache-2.0
OpenAlex works API	Dataset (AI research volume)	CC0
HN “Who is hiring?” threads via HN Algolia API	Dataset (job posts, aggregate counts)	Public API; Y Combinator terms
Greenhouse, Lever, Ashby, Workable job-board APIs	Dataset (public job postings)	Employers' public APIs; every posting links back
Himalayas, Jobicy, Remotive, Remote OK, Arbeitnow APIs	Dataset (remote job postings)	Public APIs, credited and linked as their terms ask
Wikidata (QLever SPARQL), Hipo university-domains-list, GeoNames	Dataset (universities, locations)	CC0, MIT, CC BY 4.0
O*NET-SOC occupation codes	Dataset (role catalog)	CC BY 4.0
WEF Future of Jobs 2023 and 2025, ILO Working Paper 140, Hays 2025	Dataset (cited claims)	Cited with attribution; ILO CC BY 4.0
Lesson sources (OpenStax, Microsoft “for Beginners”, The Carpentries, UK and US government material, Wikipedia)	Dataset (lesson readings)	CC BY 4.0, MIT, OGL v3.0, US public domain, CC0, CC BY-SA 4.0
30 gift photos from Wikimedia Commons (new in Phase 2)	Dataset (images)	CC0, CC BY, CC BY-SA, public domain; each credited on the Rewards page
Google Sign-In, GitHub OAuth, Sign in with Apple (new in Phase 2)	Service APIs (sign-in)	Each provider's developer terms

All are open-source, open data or public APIs, declared and attributed in the repository README (“Data and AI disclosure”). AI-assisted development tools are declared in the Technical & Tools Declaration.

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

Almat Sakiidulla

30 October 2026